

HERMANN PAUWELLS RIVERA

Software Engineer | C++ · Audio/DSP

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PROFILE

Software engineer focused on C++ and audio/DSP (JUCE), with full-stack experience in TypeScript, React/Next.js and Node.js; final semester of a B.S. in Computer Science and Technology at Tecnológico de Monterrey, available full-time from January 2027. I built a VST3 synthesizer with automated tests, a live online store for a real business, and Android and Windows apps for my own startups.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript/JavaScript, Kotlin, SQL, Swift, C#
Audio/Native: JUCE, VST3, DSP, CMake, pluginval, FFmpeg
Web & Apps: React, Next.js, Node.js, Express, Supabase, Jetpack Compose, Electron
Tools: Git, GitHub Actions, CI, Unity

EXPERIENCE

Full-Stack Developer (freelance), Juguetería El Arbolito Jun 2026 to present

- Built and shipped to production the online store and admin panel for a single-branch toy store with Next.js, TypeScript and Supabase (catalog, cart, checkout, backend).
- Integrated Mercado Pago payments and an LLM chatbot (OpenRouter); prepared the load of the 2,395-product catalog exported from the point of sale (Eleventa) and the online/in-store inventory sync (in progress).

SELECTED PROJECTS

folk-park Personal project · C++20, JUCE 2026

- Wavetable VST3/Standalone synthesizer in C++20 and JUCE; deterministic MIDI generation and real-time DSP architecture.
- Automated suites (16/16 suites, 19/19 UI contracts) and validation with pluginval strictness 5.

Lumina Reservations (WorkHub MTY) Academic team project · Node/Express, SSE, React 2026

- Implemented the reservation backend in Node/Express (services, repositories, controllers) and real-time occupancy over SSE.
- Built the floor map and the reservation flow in React/Vite (dashboard, filters, confirmation) with role-based access.

Other projects: HeatShield · CrediFiel Challenge (Datathon 2025) · Multi-agent warehouse simulation (RL) · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

STARTUPS (FOUNDER)

HIVEMIND Co-founder · team chat for people and AI agents · private pilot Sep 2026 to present

- Android (Kotlin/Compose) and Windows (Electron) apps over self-hosted ntfy with HMAC-SHA256 signed messages and 398 tests; per-event cost cut from 2.5 s to 20 ms at 20,000 messages; agent orchestration and context compaction with TypeSafe's Jev; Rust core in progress.

recaps-studio Co-founder · AI narrated-video factory (novel → manhwa-style video) Sep 2026 to present

- Resumable 13-stage Python pipeline with an orchestrator and status page: word-timed TTS, AI images, subtitles and FFmpeg/NVENC rendering distributed across several PCs; first full videos already published on YouTube.

Chécalo Founder · nutrition app for Mexico · in development Sep 2026 to present

- Barcode scanner that computes NOM-051 warning labels with deterministic rules checked against the official gazette, plus an explainable verdict from TypeSafe's Jev; iOS-style offline PWA and Android app (Kotlin); 53 tests.

AML consultancy Co-founder (technology) · anti-money-laundering · early stage Oct 2026 to present

- Co-founder in charge of technology at an AML compliance consultancy (LFPIORPI): website with a vulnerable-activity self-assessment and risk matrix, brochure and campaign; AML system (matrices and manuals) coming next.

EDUCATION

B.S. in Computer Science and Technology (ITC), Tecnológico de Monterrey, Monterrey Campus 2021 to 2026
Expected graduation: December 2026.

AWARDS & ACTIVITIES

- Winner, Sustainability category, at Swift Challenge Fest 2025 (Tecnológico de Monterrey), with the HeatShield iOS app.
- SwiftTec Coding Community (2025): hackathons and innovation challenges.
- Music production and sound design (FL Studio).

Spanish (native) · English (C1) · German A2 (2-month intensive course in Heidelberg, Germany)